

WESTERN NEBRASKA, NE, USA

WMO: 725660

Lat: 41.871N

Lon: 103.593W

Elev: 3945

StdP: 12.72

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24028

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -5.3	(c) 1.2	(d) -11.3	(e) 3.4	(f) -3.7	(g) -5.4	(h) 4.8	(i) 3.4	(j) 33.0	(k) 34.7	(l) 29.8	(m) 37.0	(n) 6.1	(o) 310	(p) 0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.4	96.7	65.5	93.6	65.2	90.5	64.8	70.6	86.2	68.9	84.6	67.6	83.5	10.0	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	111.8	76.0	64.3	104.7	74.4	62.7	98.8	73.3	37.0	86.2	35.4	84.5	34.1	83.0	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 28.8	(b) 25.0	(c) 21.0	(d) DB	(e) -15.1	(f) 102.1	(g) 7.8	(h) 2.1	(i) -20.7	(j) 103.6	(k) -25.3	(l) 104.8	(m) -29.6	(n) 105.9	(o) -35.3	(p) 107.5	(q) 86.4
(a) 28.8	(b) 25.0	(c) 21.0	(d) WB	(e) -15.5	(f) 75.3	(g) 7.7	(h) 4.3	(i) -21.1	(j) 78.4	(k) -25.5	(l) 80.9	(m) -29.8	(n) 83.3	(o) -35.4	(p) 86.4	(q) 86.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.5	27.9	30.4	39.4	46.7	56.7	68.1	74.9	72.4	63.1	48.9	37.2	27.5
	DBStd	19.04	11.23	11.52	10.45	9.09	8.52	7.36	5.24	5.28	8.41	9.10	9.90	11.61
	HDD50	3006	687	549	347	163	35	1	0	0	9	127	390	698
	HDD65	6433	1151	967	794	549	278	51	4	10	131	501	834	1163
	CDD50	2838	0	1	18	64	244	545	773	694	401	92	6	0
	CDD65	792	0	0	0	0	21	144	312	240	74	1	0	0
	CDH74	10542	0	0	23	101	493	2004	3697	2809	1245	166	4	0
	CDH80	4986	0	0	1	13	160	920	1952	1367	534	39	0	0
	WSAvg	9.4	9.5	10.2	10.6	11.4	10.5	9.5	8.2	7.9	8.0	9.0	8.9	9.1
	PrecAvg	15.4	0.4	0.5	1.0	1.7	2.6	2.6	2.0	1.1	1.1	0.9	0.6	0.5
(15) Precipitation	PrecMax	24.8	1.3	1.9	2.6	3.9	7.2	6.6	4.8	3.5	4.2	3.0	2.2	1.5
	PrecMin	7.7	0.0	0.0	0.2	0.3	0.3	0.6	0.1	0.0	0.0	0.1	0.0	0.0
	PrecStd	4.1	0.3	0.5	0.7	1.0	1.7	1.6	1.3	0.9	0.9	0.7	0.6	0.4
	PrecStd	4.1	0.3	0.5	0.7	1.0	1.7	1.6	1.3	0.9	0.9	0.7	0.6	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.8	66.9	77.3	81.7	90.1	99.0	101.5	98.2	94.9	84.8	73.0	61.5
		MCWB	44.4	45.8	50.7	55.4	60.8	65.8	65.7	64.1	63.9	57.3	49.6	45.0
	2%	DB	54.8	59.9	70.8	76.9	84.5	93.8	97.8	94.7	90.5	78.8	66.3	55.5
		MCWB	40.4	43.2	49.0	53.7	59.1	64.3	66.4	65.2	62.2	55.0	47.0	41.3
	5%	DB	49.7	54.1	65.0	71.6	79.6	89.8	94.5	91.6	86.7	73.0	60.7	50.0
		MCWB	38.3	40.4	46.1	51.4	57.8	64.0	66.0	65.5	61.2	52.9	44.6	38.6
	10%	DB	44.6	48.3	59.1	66.1	74.5	85.3	90.8	88.2	81.7	67.1	54.8	44.7
		MCWB	35.7	37.4	43.8	48.8	56.6	63.4	65.7	65.2	59.8	50.2	41.9	35.6
	0.4%	WB	45.1	47.2	54.1	58.3	66.2	73.0	73.3	72.3	68.6	60.1	50.7	45.4
		MCDB	58.4	64.7	71.1	76.5	81.0	89.6	89.2	87.5	86.7	78.3	69.7	59.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.5	43.7	49.6	54.6	62.1	68.8	70.9	69.9	65.7	56.5	47.8	41.8
		MCDB	53.4	58.7	68.1	72.3	76.9	84.2	86.8	85.2	82.2	74.4	65.0	54.1
	5%	WB	38.7	41.0	47.1	52.1	59.8	66.5	69.2	68.3	63.5	53.9	45.3	38.9
		MCDB	48.6	53.1	63.0	69.4	74.2	82.7	85.2	83.5	79.7	68.7	60.2	49.1
	10%	WB	36.0	37.9	44.5	49.8	57.7	64.5	67.9	66.8	61.7	51.5	42.4	35.9
		MCDB	44.3	47.4	58.3	64.5	71.9	80.6	84.1	81.9	76.6	64.8	53.9	44.2
(35) Mean Daily Temperature Range	5% DB	MDBR	25.1	25.7	28.2	27.3	26.3	28.5	29.4	29.7	30.8	29.1	27.6	25.2
		MCDBR	31.8	34.0	39.3	39.2	35.4	35.0	35.4	34.9	38.0	40.5	37.9	31.9
		MCWBR	19.9	19.5	20.7	19.6	14.6	12.8	11.2	11.5	14.2	19.6	21.1	20.0
		MCDBR	29.9	32.7	36.2	36.3	30.7	31.4	29.7	29.9	32.9	35.4	36.0	30.3
	5% WB	MCWBR	19.3	19.2	20.1	18.9	14.0	14.2	11.1	11.7	13.7	17.8	20.4	19.7
		MCWBR	19.3	19.2	20.1	18.9	14.0	14.2	11.1	11.7	13.7	17.8	20.4	19.7
(40) Clear-Sky Solar Irradiance	taub		0.248	0.258	0.280	0.310	0.333	0.340	0.369	0.359	0.322	0.282	0.262	0.250
	taud		2.580	2.546	2.513	2.432	2.439	2.445	2.401	2.415	2.498	2.579	2.555	2.551
	Ebn at Noon		296	308	310	305	298	295	285	285	291	294	286	285
	Edh at Noon		21	25	30	35	36	36	36	35	30	24	21	20
(44) All-Sky Solar Radiation	RadAvg		714	986	1377	1690	1956	2313	2280	1993	1608	1108	780	595
	RadStd		31	57	110	110	162	135	88	65	90	95	44	36

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page